UMHackathon 2026

TEAPOT

Your Best AI Business Intelligence Partner

PRD · System Analysis · QA Documentation

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01

Product Overview

Product Requirement Document

Problem Statement & Solution

The Problem

- Starting a business feels overwhelming — margins, market pain points, feasibility
- AI tools are stuck at theory level without actionable guidance
- No unified platform for idea validation, risk analysis, and strategy
- Amazing ideas buried under fear or poor money management

Teapot's Solution

- AI-powered platform bridging theory and real-world execution
- 7 specialized agents for deep, domain-specific analysis
- Multi-LLM backbone: Z.AI GLM, Google Gemini, Anthropic Claude
- Interactive chat interface + project management + doc generation

7 Specialized AI Agents

Sales Predictor

Forecasts sales trends and revenue opportunities based on business data.

Pain Point Analyser

Identifies key customer pain points and market gaps for the business.

Profit Optimiser

Recommends strategies to maximize profit margins and reduce waste.

Decision Recommendation

Provides data-driven strategic decisions and next-step guidance.

Risk Identifier

Evaluates risks using structured analysis and mitigation strategies.

Scenario Simulator

Simulates best/worst/expected-case business scenarios for planning.

Resource Optimizer

Allocates budgets, time, and personnel for maximum efficiency.

Custom Agent Builder

Admins can create custom agents with configurable LLM backends.

Target Audience & Scope

Intended Users

- Entrepreneurs & aspiring business starters
- Small-to-Medium Business Owners
- Startup Founders & corporate innovation teams
- Freelancers needing structured business insight
- Business students seeking practical scenario practice

Key Features (In-Scope)

- Conversational AI interface with multi-agent consultation
- Project workspace — save, track & revisit analyses
- Role-based Admin dashboard with token monitoring
- Document generation: PDF & PowerPoint export
- OAuth 2.0 login (Google / GitHub) + JWT auth

02

System Architecture

System Analysis Documentation

Technology Stack

Frontend

Next.js (React Framework)
Responsive chat UI, project management, and admin dashboard.

Backend

FastAPI (Python)
RESTful APIs, AI agent orchestration, authentication & routing.

Database

PostgreSQL (NeonDB)
User data, project history, AI outputs, chat logs & system logs.

Cloud / Deployment

Render
Automated CI/CD, scalable hosting, continuous integration.

LLM Providers

Z.AI GLM · Google Gemini · Anthropic Claude
Multi-model fallback for resilience.

Auth & Security

OAuth 2.0 (Google/GitHub)
JWT tokens, encrypted API key storage, role-based access.

System Data Flow

- 1 · User inputs a business idea or query via the conversational chat interface
- 2 · Backend performs intent analysis & routes to relevant AI agents
- 3 · Selected agents (e.g., Risk Identifier, Profit Optimiser) process the request independently
- 4 · Outputs are aggregated, validated, and synthesized into a unified recommendation
- 5 · Results stored in PostgreSQL (project tracking, history, AI outputs)
- 6 · Final insights returned as structured text or generated documents (PDF / PPT)
- 7 · Iterative refinement — user feedback loops enable strategy improvement over time

Non-Functional Requirements

Quality	Requirement	Implementation
Scalability	Multiple concurrent users without degradation	Render cloud + efficient API routing to LLMs
Reliability	Consistent AI responses, no breakdowns on API failure	Multi-LLM fallback (GLM → Gemini → Claude) + retry logic
Token Latency	Responses within 3–5 s (p95 target)	Async calls, response streaming, prompt optimization
Cost Efficiency	Controlled token budget per session	Caching (24h), summarization, selective agent triggering
Maintainability	Easy updates to agents, models, backend	Modular architecture + CI/CD via Render

External Dependencies & Risk

Tool / Service	Purpose	Risk Level
Z.AI GLM API	Core LLM — structured reasoning & multi-step prompting	High — fallback models mitigate
Google Gemini API	Secondary LLM for enhanced reasoning & fallback	Medium — multi-model redundancy
Anthropic Claude API	Conversational quality & safety-aligned outputs	Medium — fallback available
Render (Cloud)	Backend hosting, deployment scaling, uptime	Medium — monitoring + redeployment
PostgreSQL (NeonDB)	Persistent data storage for all system entities	High — managed service + backups
OAuth (Google/GitHub)	Secure user authentication and login	Low — standard login fallback

03

QA & Testing

Quality Assurance Testing Documentation

Test Strategy & Environments

Unit Tests

PyTest framework
Mocked DB & Cloudinary API
100% pass rate for happy & negative cases

Integration Tests

Full agent-to-LLM workflow
Real API calls to DB & endpoints
Test API keys for LLM providers

Regression Tests

Run on every merge to main
Min 90% pass rate across all agents
100% for critical authentication tests

CI/CD Pipeline

GitHub Actions — triggers on every push
Linting → Unit Tests → Coverage
Blocks PR merge if thresholds fail

Performance Tests

Locust / Postman load testing
50 concurrent users
Target: < 800ms API response

AI Output Testing

Min 80% acceptable prompt/response pairs
Fallback & hallucination validation
Boundary & oversized input handling

CI/CD Release Thresholds

Gate	Check	Requirement	Project Result
Integration (→ Main)	Automatic Build	0 Errors	✓ Passed
Integration (→ Main)	Unit Tests	100% Pass Rate	✓ 100%
Integration (→ Main)	Code Quality (Lint)	0 Linting Errors	✓ 100%
Integration (→ Main)	Test Coverage	≥ 81.2%	✓ 100%
Deployment (→ Prod)	Regression Tests	≥ 90%	✓ 98%
Deployment (→ Prod)	AI Output Pass Rate	≥ 80% of prompts	✓ 95%
Deployment (→ Prod)	Critical Bugs (P0/P1)	Zero	✓ Clean
Deployment (→ Prod)	API Performance	< 800ms	✓ ~500ms

Test Case Highlights

TC-01 · Happy Path

Full user journey: login → choose agent → chat → upload file → pin AI reply to dashboard.
Status: ✅ Success

TC-02 · Negative Case

Avatar upload with invalid file type (.txt). System blocks upload & displays clear error message.
Status: ✅ Success

TC-03 · Performance

50 concurrent GET requests to payment API endpoint.
Avg response: 1328ms (target < 800ms).
Status: ❌ Failed — optimization needed

Risk Assessment (Sample)

- Risk: User selects an AI model not available in backend → Score 15 (L:5 × S:3)
- Mitigation: Replace free-text model input with a validated dropdown selection
- Testing: Mock user accounts attempt to access restricted models; verify system blocks access

- Key Risks Tracked Across the System:
- • API rate limits causing agent timeouts → fallback LLM chain activated
- • Hallucinated business recommendations → post-processing validation + retry logic
- • Performance degradation under load → async calls + response length caps
- • Data loss from DB failure → managed PostgreSQL with automated backups

04

AI Design & Prompting

Model Selection, Strategy & Failure Handling

Prompting Strategy

Multi-Step Agentic Prompting

- Complex business problems decomposed into domain-specific sub-tasks
- Each agent uses structured role-based instructions + context-aware prompts
- Intermediate agent output passes as input to the next agent
- Enables layered analysis: feasibility → risk → profit → strategy

Input Handling & Context

- Max input: ~3,000–8,000 tokens depending on model context window
- Oversized inputs chunked by paragraph/topic boundaries
- Hierarchical summarization condenses chunks before agent processing
- Unprocessable inputs prompt user to refine or restructure

Fallback & Failure Handling

- Step 1 — Post-processing validation: each agent output checked for relevance & format
- Step 2 — Auto-retry with adjusted prompt (stricter context, clearer constraints)
- Step 3 — LLM switch: if retry fails, fall back to next model (GLM → Gemini → Claude)
- Step 4 — Graceful degradation: partial structured response + user guidance to refine input
- Deployment strategy: Grayscale rollout (5% → 50% → 100%) with A/B testing on prompt structures
- Emergency Rollback (ER): triggered at >20% API failure rate, reverts to Golden Release within 24h

Thank You

Teapot — Empowering entrepreneurs with AI-driven business intelligence.

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